
Which Eviction Policy Should an LLM Cache Use? A Seven-Policy Study Across Workloads, Capacities, and Encoders

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Abstract

Semantic caches reduce LLM serving cost by answering queries whose embeddings fall within a similarity threshold of a cached entry, and many eviction policies have been proposed for them, from classical replacement heuristics to policies that reason about embedding geometry. These proposals have not been compared on a common harness, and published claims conflict. We evaluate seven policies (FIFO, LRU, LFU, ARC, GDSF, SISO, and a semantic redundancy policy) on identical query streams across three workloads, three cache capacities, and two embedding encoders, with three seeds per cell, and audit the semantic validity of the resulting hits with an LLM judge. Policy choice is close to irrelevant: no policy beats LFU by more than 0.05 pp in any of eighteen cells, because the cache’s own admission rule deduplicates its contents and starves geometry-aware policies of signal. Two other design decisions dominate policy choice. Similarity thresholds do not transfer across encoders, and reusing one silently yields a perfect but meaningless hit rate; and most hits at the standard threshold are not semantically equivalent queries, collapsing quality-adjusted hit rates from 51–60% to 1.1–2.2% on conversational workloads. Deployments should default to LFU and spend their effort on threshold calibration and hit auditing.

1 Introduction

A semantic cache intercepts an LLM query, embeds it, and serves a stored response if a previously answered query lies within a distance threshold, bypassing the model entirely [1]. Of the design decisions such a cache requires, eviction has attracted the widest range of competing proposals: production systems default to classical heuristics such as LRU and LFU, hybrids such as ARC [11] and GDSF [2] blend recency and frequency signals, and a newer family argues that a semantic cache should instead preserve semantic coverage, evicting entries whose neighbors make them redundant. This family includes SISO [8], designed specifically for LLM serving, bandit formulations [10], learning-augmented replacement for agent retrieval buffers [14], and semantic-aware eviction for prefix caches [4].

Each proposal was evaluated in the setting its authors chose, and the published claims conflict: Sun et al. [14] report LRU and LFU consistently losing to a naive FIFO baseline on semantic workloads, while serving-oriented papers treat frequency signals as clearly valuable. A practitioner has no basis for choosing. We resolve the question by running the policies against each other: seven policies on a common harness, so that every policy sees an identical query stream, index, similarity threshold, and measurement protocol, across three workloads spanning the redundancy spectrum, three cache capacities, and two embedding encoders. Because hit rate cannot distinguish a useful cache from a permissive one, we additionally audit sampled hits with an LLM judge [15].

The comparison yields a deflationary but useful answer. The policy axis is nearly inert because the cache’s admission rule structurally suppresses the signal that semantic policies consume (Section 4). The decisions that actually move outcomes, by orders of magnitude rather than hundredths of a point, are threshold calibration across encoders and admission quality (Sections 3.2 and 3.3).

2 Study design

Harness. Queries stream in temporal order through a fixed-capacity cache. A query whose nearest cached embedding lies within an L_2^2 threshold of 0.90 (cosine ≥ 0.55 for unit-norm vectors) is a hit and is served without insertion; a miss inserts the query, evicting under the policy being tested when the cache is full. Policies observe only insert, access, and evict events. The first 30% of each stream is warmup; hit rates cover the remaining 70%. Each cell runs seeds {42, 123, 456}; the maximum hit-rate standard deviation across all 126 policy cells is 3.1×10^{-4} , so orderings are essentially deterministic.

Policies. FIFO (arrival order; the weakest functioning control, included because Sun et al. [14] report it beating LRU/LFU), LRU, LFU, ARC [11] (with ghost hits detected semantically, since cache ids are slot-unique), GDSF [2] (which reduces to LFU-with-aging on unit-norm fixed-dimension embeddings), SISO [8] (an online single-pass adaptation of the LLM-serving-specific cluster-centroid design), and a semantic redundancy policy that scores entries by the ratio of neighborhood redundancy to recency-frequency utility, with Laplace smoothing $\mu = 0.1$ so it degrades to inverse-utility ordering when the redundancy signal is absent.

Workloads and encoders. 100K-query temporal streams from LMSYS-Chat-1M [16] (diverse real conversations), Quora Question Pairs [7] (high-redundancy questions), and MOSS [13] (templated synthetic instructions; a saturation control), at cache capacities of 10%, 20%, and 30% of the stream. All queries are embedded with all-MiniLM-L6-v2 [12], and the full matrix is replicated under gte-base [9] after per-encoder threshold recalibration (Section 3.2).

Judged hit quality. For the 10%-capacity MiniLM runs we log every served hit and stratified-sample 1,000 hits per dataset-policy cell across distance quantiles, yielding 13,669 unique query pairs. A local Llama-3.1-8B judge [6] answers whether a correct, complete answer to the cached query would also correctly and completely answer the new one. On 1,931 pairs re-judged by an independently hosted runtime, agreement is 98.55% (Cohen’s $\kappa = 0.826$).

3 Results

3.1 Policy choice is close to irrelevant

Figure 1 shows the full matrix. Across all eighteen dataset, capacity, and encoder cells, no policy beats LFU by more than 0.05 pp; the largest margin anywhere is 0.0409 pp, a handful of queries out of $\sim 70K$. ARC matches LFU to within 0.01 pp in every cell, its adaptation parameter starved by the same signal sparsity that disables the semantic policy. GDSF behaves as its reduction predicts. The semantic policy tracks LFU while costing $5.8\text{--}8.2\times$ LRU’s per-query time.

The bottom of the ranking separates the policies more clearly. FIFO never beats LRU and ranks sixth or seventh of seven in seventeen of eighteen cells, so the FIFO-beats-LRU/LFU ordering reported for agent memory buffers [14] does not reproduce on serving traces; their buffers see almost no repeat access, whereas serving streams repeat popular queries often enough for frequency counts to matter at tight capacity. SISO, the one policy designed specifically for LLM caches, is indistinguishable from this FIFO control on the sparse workloads: across the twelve LMSYS and QQP cells the two differ by 0.096 pp on average (at most 0.254 pp, against a seed standard deviation of 0.006 pp), and SISO is measurably worse than FIFO in nine of the twelve, while costing $3.1\text{--}3.5\times$ LRU’s per-query time. A policy that consults embedding geometry recovers less than one that ignores the embeddings entirely.

Every remaining gap closes as the cache grows. LFU’s lead over LRU falls from 1.5 pp (LMSYS) and 2.5 pp (QQP) at 10% capacity to at most 0.012 pp at 30%, and the SISO and FIFO deficits shrink on the same schedule, so added capacity buys more than any replacement decision.

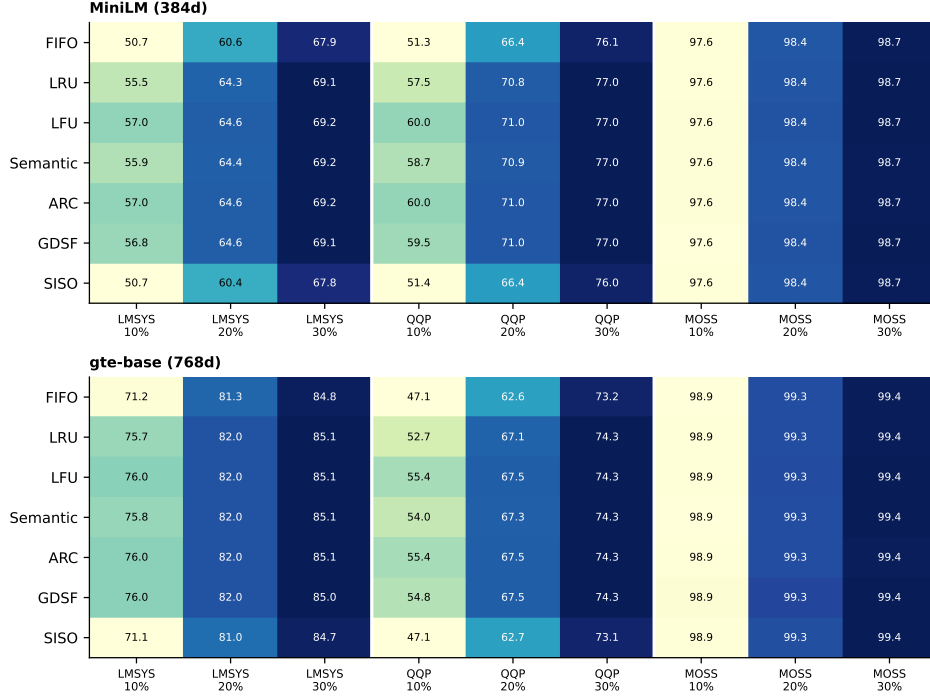


Figure 1: Hit rate (%) for all seven policies across three datasets and three cache capacities, under MiniLM (top) and recalibrated gte-base (bottom); color is normalized within each dataset block. LFU is never beaten by more than 0.05 pp, and the SISO row is nearly indistinguishable from the FIFO control.

3.2 Thresholds do not transfer across encoders

Rerunning the matrix under gte-base with the MiniLM-calibrated threshold is degenerate: gte-base is strongly anisotropic [3], with random-pair cosine similarity averaging 0.69–0.76 on our corpora, so every query clears the threshold, the measured hit rate is exactly 1.0 with zero evictions, and the experiment measures nothing. A deployment that swaps encoders without recalibrating silently degrades into serving arbitrary responses at a perfect measured hit rate. After anchoring each distance parameter at the same quantile of the new encoder’s similarity distribution, the policy ordering of Section 3.1 replicates exactly, but absolute hit rates do not transfer: LFU moves by +19.0 pp on LMSYS and −4.6 pp on QQP, differences that are large and not even direction-consistent.

3.3 Most hits are not equivalent queries

Figure 2 shows the judged audit. At the standard threshold, judged-equivalent fractions are 3.1–3.9% on LMSYS and 2.1–3.1% on QQP, collapsing raw hit rates of 51–60% to quality-adjusted rates of 1.1–2.2%; even on saturated MOSS, only about a quarter of hits are equivalent. Within every dataset the policies’ judged fractions are indistinguishable from binomial noise at $n=1,000$ per cell, so quality adjustment also erases the residual ordering the policies were competing over. Near-zero embedding distance does not rescue precision: LMSYS pairs in the nearest distance decile (median $L_2^2 = 0.10$) are judged equivalent only 8.9% of the time, because MiniLM saturates on long templated prompts that share boilerplate but differ in payload. The decisive design choice in a semantic cache is therefore admission rather than eviction; verified caching reaches the same conclusion from the design side [5].

4 Why the policies converge

The convergence has a structural cause. A query within the threshold of a cached entry is served and never inserted, so the cache accumulates precisely the residue of mutually distant queries; the

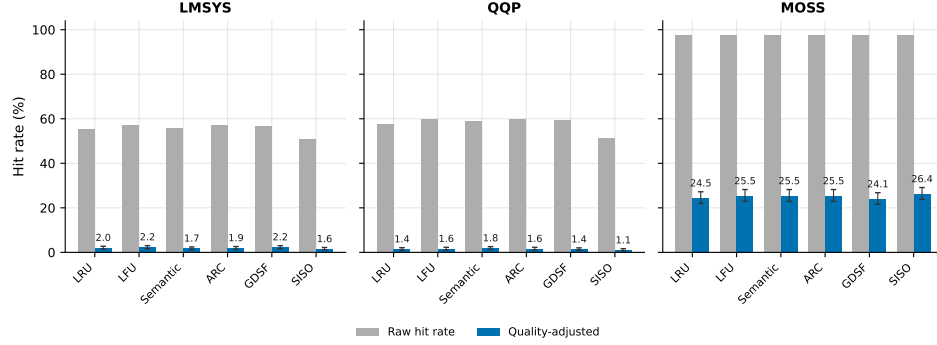


Figure 2: Raw vs. quality-adjusted hit rate per dataset and policy (10% capacity, MiniLM, Wilson 95% CIs). Quality adjustment collapses LMSYS and QQP hit rates by more than an order of magnitude and erases the differences among policies.

denser the workload, the more aggressively admission filters its redundancy out. Profiling the cache contents confirms this inversion: QQP, the most redundant workload we tested, produces the sparsest cache contents (mean within-threshold neighbor fraction 2.1×10^{-5} , versus 4.1×10^{-4} for LMSYS and 2.5×10^{-3} for MOSS). Even the densest value sits $40\times$ below the semantic policy’s smoothing constant $\mu = 0.1$, so its scoring function reduces to inverse-utility ordering on every dataset, and ARC’s semantic ghost hits and SISO’s cluster mass starve for the same reason. Because the neighbor threshold equals the hit threshold, a semantic cache deprives its own eviction policy of the signal it is built to exploit, on any workload.

5 Guidance and limitations

Deployments should default to LFU, which is within 0.05 pp of the best policy everywhere we measured and among the cheapest to run. Capacity should be provisioned before replacement is tuned, since every policy gap collapses as the cache grows. Before adopting a semantic policy, a deployment should measure cache-content density under its intended threshold, a cheap check that every workload we tested fails by at least $40\times$. The effort recovered from eviction tuning belongs in per-encoder threshold calibration and judged spot-audits of sampled hits, the two decisions that changed outcomes by orders of magnitude in this study. Our study simulates responses rather than storing model outputs, judges query equivalence rather than answer reusability, uses compact sentence encoders, and audits one operating configuration covering six of the seven policies; the harness, artifacts, and full-length analysis are available for extension.

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A Judge rubric and validation

The judge is Llama-3.1-8B-Instruct served locally via Ollama at temperature 0 with a 16-token completion cap. The system prompt casts it as a strict evaluator of semantic cache hits that must judge meaning, not wording, and answer NO if reusing the original answer could mislead or omit something the new question specifically asks for. The user prompt presents both queries and asks: “Would a correct, complete answer to the ORIGINAL question also be a correct and complete answer to the NEW question? Reply with exactly one word: YES or NO.” 13,667 of 13,669 pairs parse cleanly. We judge query equivalence rather than grounding on generated responses because QQP provides no responses and a response-grounded protocol would reintroduce the embedding pipeline’s own similarity signal into its evaluation. The 1,931-pair cross-runtime overlap (local Ollama vs. Groq’s hosted llama-3.1-8b-instant, identical rubric and decoding) yields 98.55% raw agreement, $\kappa = 0.826$; all 28 disagreements concentrate at close distances (median L_2^2 of 0.236 vs. 0.606 for agreements), i.e., genuinely borderline pairs.

B Per-encoder recalibration

MiniLM’s hit threshold of $0.90 L_2^2$ (cosine 0.55) sits at its median nearest-neighbor similarity over the 100K-query corpora. For gte-base every distance-bearing parameter is re-anchored at the same quantile of the gte similarity distribution: the hit threshold becomes $0.30 L_2^2$ (cosine 0.85, the gte nearest-neighbor median), the semantic policy’s redundancy threshold moves with it, SISO’s clustering threshold moves from cosine 0.86 to 0.94, and ARC’s ghost-match threshold tightens from 0.20 to $0.10 L_2^2$ because at 0.20 ghost hits fired on 47–89% of merely topical gte neighbors. All non-geometric hyperparameters are unchanged.

C Reproducibility

Every run writes a JSON manifest recording its command line, config hash, package versions, Git commit, and host hardware. Eviction cells use seeds {42, 123, 456}; the judge sample derives from the seed-42 runs. The semantic policy’s hyperparameters are $\alpha = \beta = 1.0$, $\mu = 0.1$, $\varepsilon = 10^{-9}$, and a neighbor-sample cap of 1,024. Per-query overhead relative to LRU at 10% capacity: semantic $4.3\times$ (LMSYS) and $3.8\times$ (QQP); SISO $3.5\times$ and $3.1\times$; on saturated MOSS, where evictions are rare, overheads collapse toward $1\times$.